

Sales Performance Report

Superstore 2015-2018: overall performance, promotion effectiveness and customer behaviour

Data Analyst portfolio project | SQL, Python, dashboard

Executive summary

Growth is healthy, but discounting and a stalled customer base put it at risk

\$2.30M

Total sales

\$286K

Profit (12.5% margin)

+51%

Sales growth 2015 to 2018

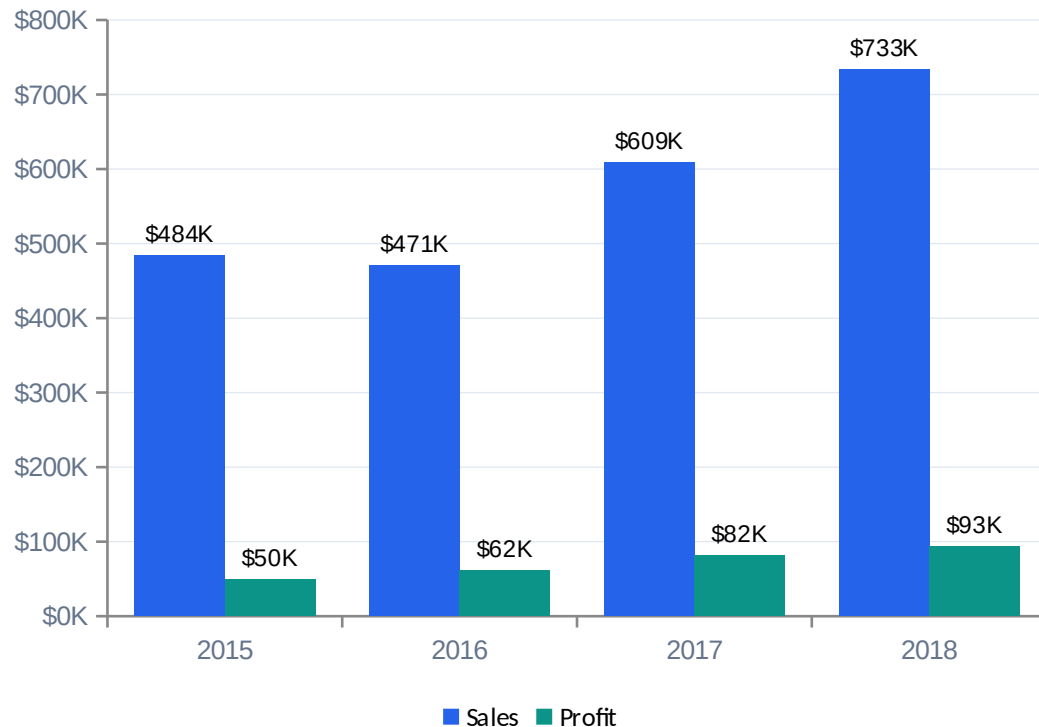
-\$135K

Lost on >20% discount lines

- Discounts above 20% are 14% of order lines but erase 32% of the profit earned elsewhere, with no gain in units per order.
- Tables, Bookcases and Supplies lose money; Copiers grows fastest with a 37% margin.
- New customers fell from 136 (2016) to 11 (2018): growth relies on a saturated base of about 790 customers.

Sales grew 51% since 2015; margin peaked in 2017

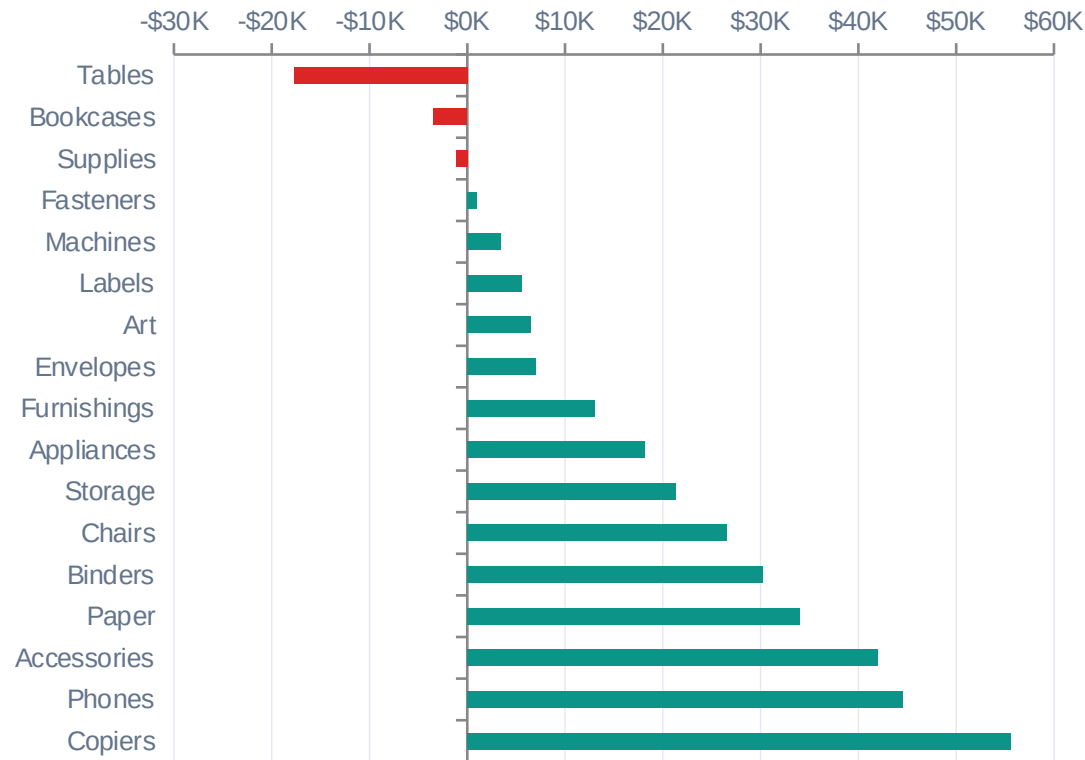
Sales and profit by year (USD)



- 2016 dipped 2.8%, then +29% (2017) and +20% (2018).
- Margin: 10.2% (2015), 13.4% (2017), 12.7% (2018).
- Orders up 74% but average order value down from \$500 to \$435.

Three sub-categories lose money

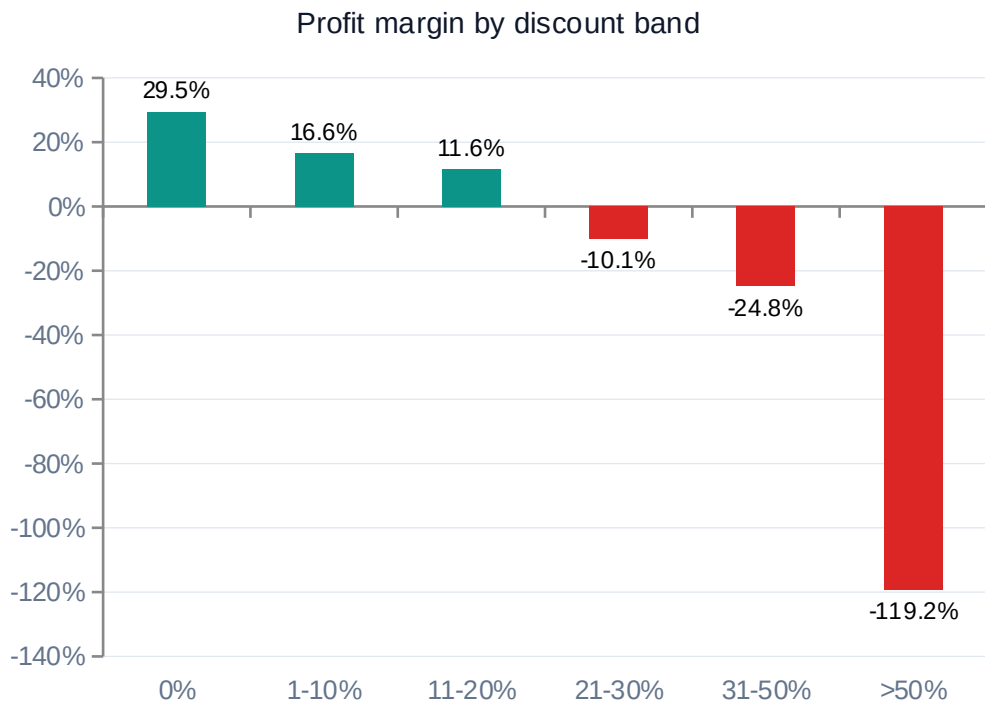
Total profit by sub-category, 2015-2018 (USD)



- Tables: \$207K sales but -\$18K profit (-8.6% margin); 55% of lines discounted over 20%.
- Copiers: sales CAGR +80% with a 37% margin.
- Machines: sales CAGR -11% and loss-making in 2018.
- Paper, Labels and Envelopes: 42-44% margins on small revenue; bundling candidates.

Discounts above 20% destroy margin without lifting volume

Profit margin and average quantity by discount band



- Margin 29.5% at no discount, 11.6% at 11-20%, then negative: -10%, -25%, -119%.
- Average quantity per line stays between 3.7 and 4.0 in every band.
- Lines above 20%: 14% of volume, -\$135K profit, equal to 32% of the \$422K earned at 20% or less.
- Discount is a proxy for promotions; the data shows association, not causation.

Central's low margin is a discount problem

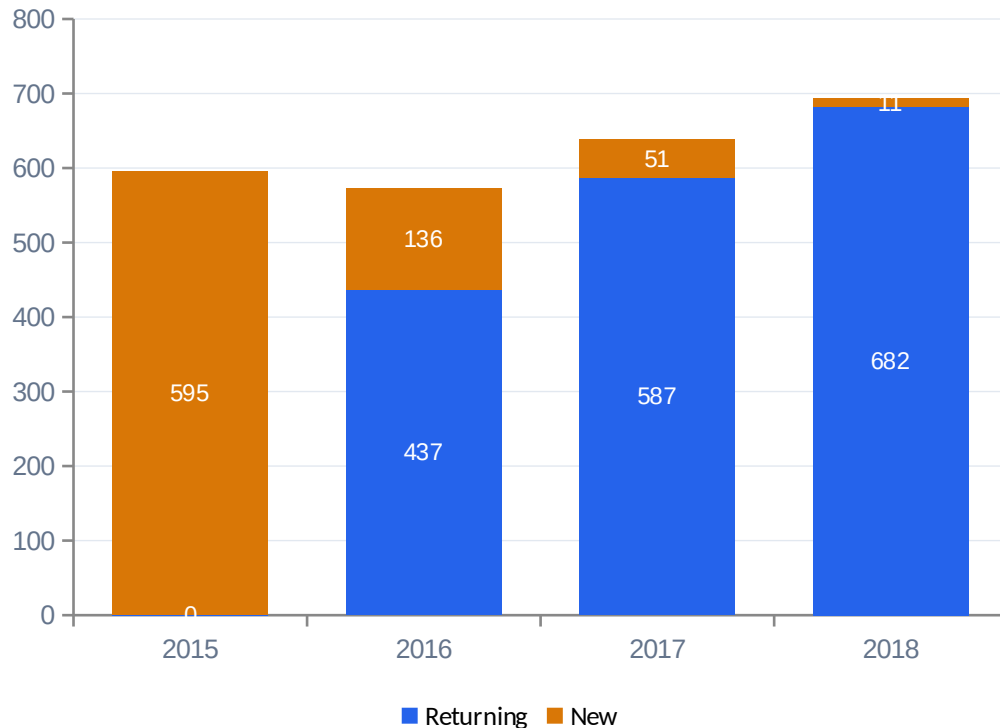
Region view: margin with and without lines discounted over 20%

Region	Margin	Avg discount	Lines >20% disc.	Profit lost >20%	Margin if <=20%
Central	7.9%	24.0%	27.7%	-\$52K	24.8%
East	13.5%	14.5%	16.2%	-\$42K	25.1%
South	11.9%	14.7%	10.6%	-\$25K	22.3%
West	14.9%	10.9%	3.7%	-\$16K	17.6%

- Central has the lowest margin (7.9%) and the heaviest discounting (24% average, 28% of lines above 20%).
- Excluding lines above 20%, Central's margin (24.8%) is on par with East (25.1%): the gap comes from discount policy, not demand.

Growth comes from existing customers

Active customers per year: new vs returning (2015 is the first year in the data)



- New customers: 136 (2016), 51 (2017), 11 (2018).
- Retention rose from 73% to 87%; 98% of customers order at least twice.
- The top 20% of customers generate 48% of sales.
- RFM: Champions are 24% of customers and 40% of sales; At Risk plus Lost are 25% of customers and 8% of sales.

Recommendations

Ordered by expected profit impact

1

Cap discounts at 20%

Require approval above the cap. Deep discounts show no volume benefit.

2

Fix Tables, Bookcases, Supplies

Re-price or restructure; over half of Tables lines carry deep discounts.

3

Review Central region pricing

Margin gap versus East and West is a pricing-discipline issue.

4

Grow winners, bundle high-margin lines

Copiers, Accessories, Appliances; add Paper, Labels, Envelopes to orders.

5

Restart customer acquisition

Win-back for At Risk, loyalty tier for Champions.